VISHNU K

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KEY ACHIEVEMENTS

- Built a multi-tenant AI SaaS platform for hotel social media marketing at FotoBubbles Tech (May 2026 – Present); creating a hotel campaign takes under 5 minutes in the tool instead of hours
- Wrote that platform's test suite: 200+ automated tests (Vitest, pytest, Playwright), a 90% coverage threshold enforced in CI, and mutation testing with Stryker and cosmic-ray
- Built Touchline AI, an evidence-checked football analyst (tool-calling agent, RAG, Postgres/pgvector) on StatsBomb World Cup 2022 data, deployed on Render and Vercel
- Top 50, VISA 24hr AI Hackathon @ IIT Madras, 2025 (national selection)
- 480 organic Chrome Web Store installs (Trust Layer extension), 5.0★ rating, users in US, India & Germany
- IEEE research paper under review: Thermal Anomaly Detection using CNN + Bi-LSTM with self-attention on 895 IR images

EDUCATION

B.E. – Artificial Intelligence & Machine Learning | BNM Institute of Technology, Bengaluru *2023 – Expected May–Jun 2027*
CGPA: 8.35 (8th semester)

Pre-University (PCMC): The National PU College, Jayanagar – 80% | Apr 2023

INTERNSHIP EXPERIENCE

AI/ML Intern | FotoBubbles Tech Pvt. Ltd. *May 2026 – Present*

- Built a multi-tenant AI SaaS platform for hotel marketing (Next.js 15, TypeScript, Prisma, PostgreSQL, Python/FastAPI services), owning the work from architecture to UI.
- Wrote 200+ automated tests: Vitest and pytest for logic and API behaviour, Playwright for end-to-end user flows. A 90% coverage threshold is enforced, and pre-commit/pre-push hooks and GitHub Actions run the checks.
- Used mutation testing (Stryker, cosmic-ray) to check that the tests actually catch bugs. Before pushing a fix, I confirm its new regression test fails when the fix is removed.
- Tested the develop build end to end on a real property (22 areas, 57 photos) and filed the bugs I found in ONES, each with repro steps, expected vs actual result, severity and root cause down to the file and line. Examples: distinct drone photos dropped as "duplicates" because of how filenames were normalised, and AI assigning photos to the wrong area.
- Fixed bugs from that test and from code review in pull requests, each with tests for the failing case. Examples: a status shown as "unavailable" while a background job was still retrying, and date formatting that broke across timezones.
- Designed multi-tenant data isolation using Postgres row-level security with separate database roles for app, auth and platform. Built background jobs on a pg-boss outbox with idempotency keys so a job is not processed twice.
- Integrated GPT-4o and GPT-4o Vision with JSON-schema validation to tag photos (quality, tone, best use) and write copy for 5 channels (Instagram, WhatsApp, Google Business, Meta Ads, short caption). An AI-spend ledger caps cost.
- Wrote a deterministic image-composition algorithm in Python (Pillow) using a 4×4 grid saliency score (edge density, local contrast, hero-zone protection) to render 1080×1080, 9:16 Story and carousel creatives without generative image models.
- Added OpenTelemetry traces, metrics and trace-correlated logs exported to Grafana Cloud, with alerting and an AI-spend dashboard.

AI/ML Intern | Equinoxis Pvt. Ltd. *Jun 2025 – Jul 2025*

- Prototyped an AI-assisted video generation workflow with the Google Gemini API in Python/Jupyter, tuning prompts and model parameters for output quality.
- Evaluated Gemini's multimodal capabilities for a generative media use case.
- Wrote reproducible notebooks covering Gemini API request handling, response parsing and output evaluation.

PROJECTS

Touchline AI – *FastAPI, Postgres/pgvector, OpenAI, Next.js* | **GitHub** *2026*

- Football analyst over five StatsBomb 2022 World Cup matches. An agent with 8 tools (xG, PPDA, heatmaps, minutes played, 360 freeze-frame shot context, RAG search) must cite match event ids; a validator checks every quoted number against the tool results and flags what it cannot verify.
- 130 API tests, Alembic migrations, Docker, rate limits and a daily spend cap. A live eval on 9 golden questions passed 9/9 after one fix (a small smoke test, not a benchmark). API on Render's free plan, web on Vercel.

Automated Brain Tumor Detection – *PyTorch, OpenCV, Vision Transformer* *2024–2025*

- Hybrid CNN + Vision Transformer model for MRI-based multi-class tumor classification with multimodal fusion (T1/T2/FLAIR), Grad-CAM explainability, and radiomics integration. ~98% accuracy, 0.99 ROC-AUC.

CREDA – *Skill Verification Platform, GitHub API, AI Assessment* *2025–Present*

- AI-powered skill assessment platform that analyzes resumes and GitHub repos to verify genuine coding experience vs AI-dependent work, generating portable Skill Passports with a Trust Score. 100 active users, 86 AI interviews conducted.
- Features anti-cheat detection, tiered model fallback, Redis/BullMQ job queues, and Supabase RLS.

Trust Layer – *Chrome Extension, NLP, Privacy Analysis* *2024*

- AI-powered Chrome extension scoring website privacy risk 1–10 using real-time NLP analysis of privacy policies. 480 organic Chrome Web Store installs, 5.0★ rating. Users in US, India & Germany.

Thermal Anomaly Detection – *PyTorch, Flask, ResNet-18, Bi-LSTM | IEEE Paper Under Review* *2024–2025*

- ResNet-18 + Bi-LSTM with self-attention for IR thermal image anomaly detection on power transformers. Grad-CAM explainability, trained on 895 real SciDB images.

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL, Java, C

Testing/QA: pytest, Vitest, Playwright (E2E), API testing (Postman), regression testing, test case design, bug reporting (ONES), coverage gating, mutation testing, Git hooks, GitHub Actions

Backend: FastAPI, Flask, Node.js, Express, Prisma, Alembic, REST API design, pg-boss/BullMQ job queues, better-auth, Redis, Webhooks, WhatsApp Cloud API, web scraping (httpx, BeautifulSoup, Playwright)

Frontend: Next.js, React, Tailwind CSS, HTML/CSS

Databases: PostgreSQL (pgvector, row-level security), Neon, MongoDB Atlas Vector Search, Supabase, Firebase, MySQL

ML/CV: PyTorch, TensorFlow, OpenCV, MediaPipe, YOLOv8, Vision Transformers, CNN/Bi-LSTM, Grad-CAM, Scikit-learn

LLMs/AI: OpenAI GPT-4o, Google Gemini, Perplexity, OpenRouter, Claude API, RAG, vector search, embeddings, prompt engineering, structured outputs (JSON schema), tool-calling agents, LLM evals, MCP, multi-provider failover

Automation/AI Tools: n8n, Manus, Make.com

Observability: OpenTelemetry, Grafana Cloud, Grafana, Tempo, Loki, Prometheus, Jaeger, SigNoz

DevOps: Docker, Docker Compose, Git/GitHub, Cloudflare R2 & Tunnel, Vercel, Render, Railway

Other: System design, ADRs, code review, Agile/ONES, no-code/low-code platforms

HACKATHONS & CERTIFICATIONS

Hackathons: VISA 24Hr AI Hackathon @ IIT Madras (Top 50) | AlgoBharat Hackseries 2025 (Semi-finalist) | BITS Pilani Hackathon 2025, Hyderabad Campus (Participant)

Certifications: Anthropic – Introduction to Agent Skills | Anthropic – Introduction to Model Context Protocol (MCP) | Infosys Springboard – Blockchain Technology, HTML & CSS3 | Postman – API Testing & Integration Essentials